

# Auditable Hierarchical Alpha Ensembling for A-Share Cross-Sectional Prediction

Strict IS/OOS Separation, Correlation-Constrained Greedy Selection, and Matrix-Level Nesting

YYB Quantitative Research Platform

Generated from frozen evidence: 2026-06-02T00:46:52

## Abstract

We study cross-sectional prediction of five-day A-share returns using a local research platform with strict in-sample (IS, 2020–2022) selection and out-of-sample (OOS, 2023) evaluation. The frozen audit compares equal weighting, ICIR weighting, and inverse-volatility shrinkage across unfiltered and correlation-constrained baskets; five neutralization regimes; and exploratory matrix-level hierarchical nesting. The best confirmatory configuration is `top_is_sharpe_n5_icir_beta` with OOS Sharpe 12.882. We additionally apply a hard portfolio screen requiring Sharpe  $> 10$  and pairwise absolute daily-PnL correlation  $< 0.5$ . The requested portfolio of ten candidates is not satisfied: only 1 candidate passes both constraints. We report this shortfall directly rather than fill the target with economically redundant variants. All headline tables and figures are generated from a single frozen JSON artifact, and exploratory nested results are separated from confirmatory evidence.

## 1 Research question and contributions

The empirical question is whether economically interpretable A-share signals can be combined into robust, low-redundancy predictors under strict temporal separation. The platform contributes: (i) an auditable IS/OOS protocol; (ii) true PnL-correlation greedy selection rather than family-name proxies; (iii) five explicit neutralization regimes; and (iv) matrix-level nesting in which a cached OOS combo matrix can become an input to a higher-level combo without reparsing its leaves. The audit also records negative results, skipped features, and missing infrastructure rather than silently excluding them.

## 2 Data and validation protocol

The local panel contains 970 trading days, 5,515 A-share securities, and 89 raw or derived fields. The target is the future five-day return. IS spans 2020–2022 and is used for ranking and weight estimation; OOS spans 2023 and is used only for evaluation. Correlation screening converts cumulative PnL to daily changes and uses absolute Pearson correlation. The confirmatory matrix is frozen before ranking OOS results. Exploratory L1–L3 nesting is explicitly labelled exploratory because its layer choices inspect OOS outcomes.

**Trading-protocol limitation.** The current research engine excludes immature listings and ST securities, but it does not yet model all execution frictions needed for a production claim: commissions, stamp duty, market impact, limit-up/down execution, T+1 constraints, and capacity. Reported Sharpe ratios are therefore signal-research metrics, not deployable net-performance claims.

## 3 Combination methods

For standardized cross-sectional signals  $z(F_i)$ , the linear combo is

$$F_c = \sum_{i=1}^N w_i z(F_i).$$

Equal weighting uses  $w_i = 1/N$ . ICIR weighting uses IS rank-IC stability,  $w_i \propto \max(0.1, \text{ICIR}_i^{IS})$ . The platform’s historical “ridge” label is retained in tables for compatibility, but the implementation is accurately described as inverse-volatility shrinkage:  $w_i \propto 1/(\sigma_i^{IS} + \text{median}(\sigma^{IS}))$ . It is not a standard ridge-regression optimizer. LightGBM is an exploratory nonlinear benchmark and is not mixed into the confirmatory table.

## 4 Neutralization ablation

The audit evaluates: no neutralization; market-cap decile demeaning; cross-sectional log-market-cap regression; cross-sectional beta regression; and joint log-market-cap plus beta regression. Reliable industry labels are not available in the current data catalog, so industry neutralization is not fabricated. Figure 1 reports mean Sharpe changes relative to the unneutralized control.

**Neutralization ablation: mean OOS Sharpe change versus no neutralization**

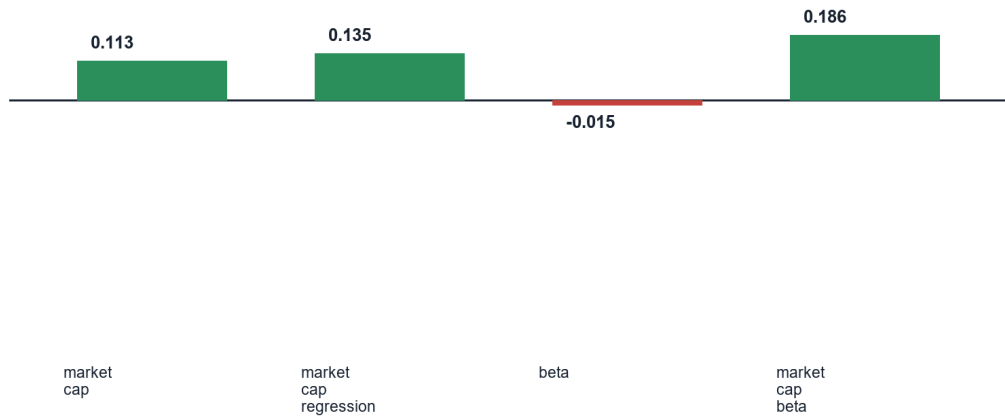


Figure 1: Neutralization ablation using frozen confirmatory baskets.

## 5 Confirmatory results

**Confirmatory OOS Sharpe ranking (IS-selected baskets, 2023 OOS)**

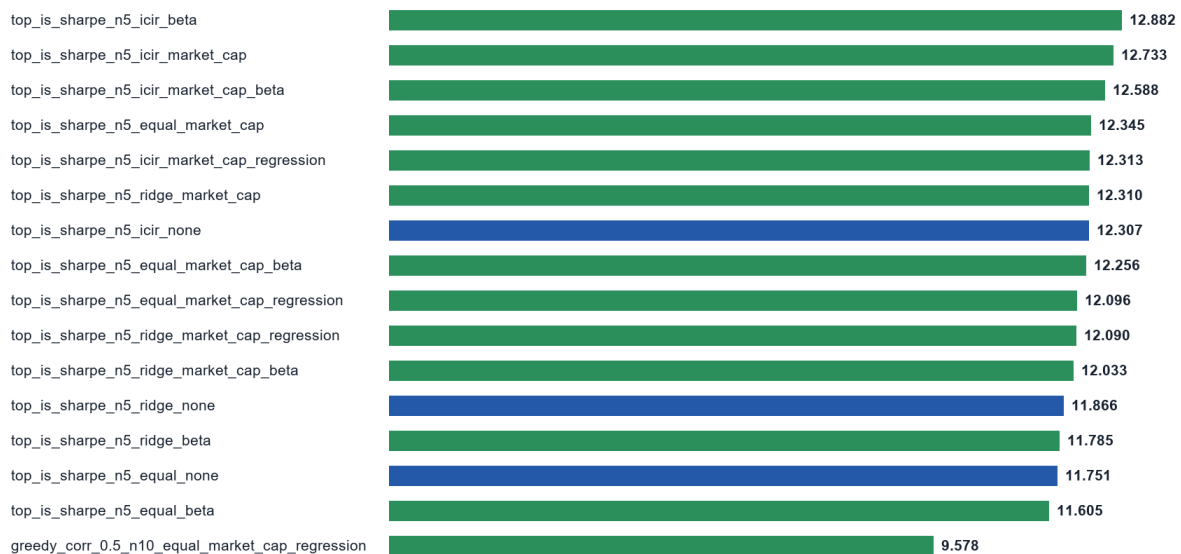


Figure 2: Top confirmatory OOS Sharpe values. Baskets and weights are selected using IS information only.

Table 1: Top frozen confirmatory configurations.

| Rank | Configuration                                | Sharpe | IC     | Turnover |
|------|----------------------------------------------|--------|--------|----------|
| 1    | top_is_sharpe_n5_icir_beta                   | 12.882 | 0.0402 | 0.2709   |
| 2    | top_is_sharpe_n5_icir_market_cap             | 12.733 | 0.0396 | 0.2695   |
| 3    | top_is_sharpe_n5_icir_market_cap_beta        | 12.588 | 0.0402 | 0.2701   |
| 4    | top_is_sharpe_n5_equal_market_cap            | 12.345 | 0.0401 | 0.2400   |
| 5    | top_is_sharpe_n5_icir_market_cap_regression  | 12.313 | 0.0395 | 0.2696   |
| 6    | top_is_sharpe_n5_ridge_market_cap            | 12.310 | 0.0406 | 0.2834   |
| 7    | top_is_sharpe_n5_icir_none                   | 12.307 | 0.0395 | 0.2698   |
| 8    | top_is_sharpe_n5_equal_market_cap_beta       | 12.256 | 0.0411 | 0.2465   |
| 9    | top_is_sharpe_n5_equal_market_cap_regression | 12.096 | 0.0400 | 0.2444   |
| 10   | top_is_sharpe_n5_ridge_market_cap_regression | 12.090 | 0.0403 | 0.2856   |
| 11   | top_is_sharpe_n5_ridge_market_cap_beta       | 12.033 | 0.0409 | 0.2858   |
| 12   | top_is_sharpe_n5_ridge_none                  | 11.866 | 0.0406 | 0.2800   |

Single-factor transportability: IS Sharpe versus 2023 OOS Sharpe



Figure 3: Single-factor IS Sharpe versus 2023 OOS Sharpe. Dispersion motivates explicit OOS isolation and economic screening.

## 6 Correlation-constrained target portfolio

The delivery target is ten factors with Sharpe  $> 10$  and pairwise absolute daily-PnL correlation  $< 0.5$ . The screen is hard: no relaxation and no duplicate PnL curves are admitted. The requested portfolio of ten candidates is not satisfied: only 1 candidates pass both constraints.

Table 2: Strict high-Sharpe, low-correlation screen.

| Rank | ID prefix    | Sharpe | IC     | Fitness | Turnover | Max corr. |
|------|--------------|--------|--------|---------|----------|-----------|
| 1    | d88b1b68-724 | 13.351 | 0.0433 | 18.316  | 0.3161   | 0.0000    |

### Strict low-correlation high-Sharpe candidates: cumulative OOS PnL

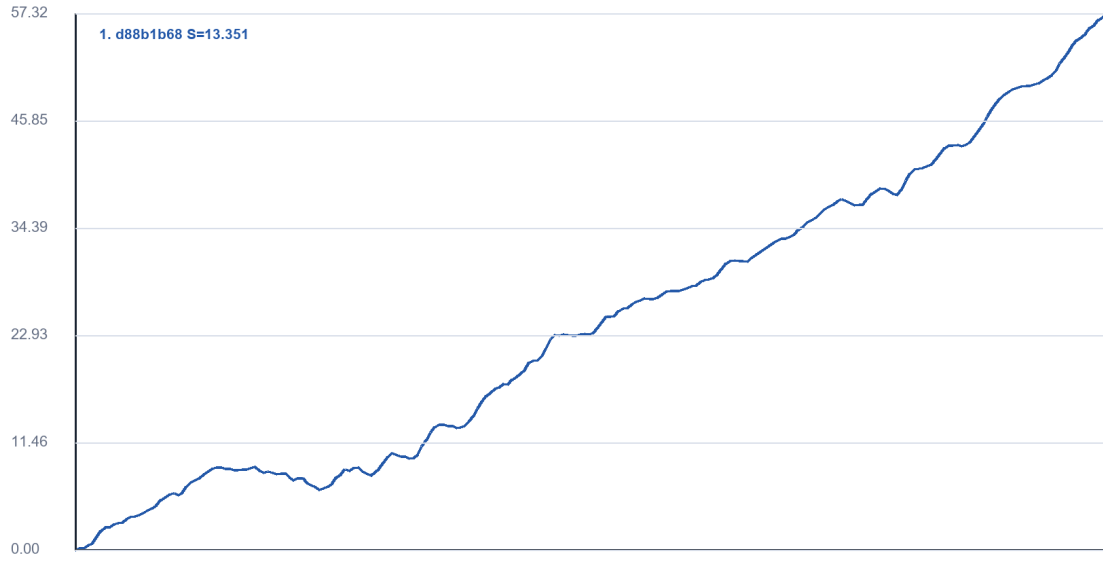


Figure 4: Cumulative OOS PnL curves for strict candidates.

### Absolute daily-PnL correlation among strict candidates

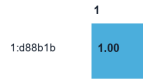


Figure 5: Absolute daily-PnL correlation heatmap for strict candidates.

## 6.1 Economic interpretation of strict candidates

Each surviving strict candidate is an ensemble rather than an atomic anomaly. The table reports its dominant economic interpretation and detected child proxies. For new nested records, child lookup follows preserved `superalpha_ref(...)` lineage recursively. For legacy cached records, the stored expression is inspected directly and missing leaf labels are reported rather than invented.

| Rank | ID prefix    | Ensemble interpretation          | Detected child proxies                    |
|------|--------------|----------------------------------|-------------------------------------------|
| 1    | d88b1b68-724 | composite cross-sectional signal | legacy composite; leaf labels unavailable |

## 7 Economic interpretation of the frozen greedy seed basket

The correlation-constrained seed basket is not presented as a causal structural model. It is an interpretable set of cross-sectional proxies used to construct candidate ensembles. Liquidity terms proxy crowding and trading frictions; return terms proxy continuation or overreaction; reversal and auction terms proxy short-lived price pressure; risk terms condition exposure on the volatility regime; and intraday terms describe price formation. Each leaf remains inspectable in the frozen manifest.

| Rank | Leaf expression                                       | Economic interpretation                                            | IS Sharpe | OOS Sharpe |
|------|-------------------------------------------------------|--------------------------------------------------------------------|-----------|------------|
| 1    | <code>-rank(log_dollar_vol)-rank(ret_120d_s...</code> | liquidity and crowdedness; medium-horizon continuation or reversal | 7.209     | 9.960      |
| 2    | <code>-rank(abs(zscore(turnover_5d)-zscore(...</code> | liquidity and crowdedness                                          | 6.327     | 3.331      |
| 3    | <code>zscore(ret_20d)-zscore(turnover_rate)</code>    | liquidity and crowdedness; medium-horizon continuation or reversal | 6.153     | 8.869      |
| 4    | <code>-rank(log_dollar_vol)-rank(ret_20d)</code>      | liquidity and crowdedness; medium-horizon continuation or reversal | 5.894     | 5.787      |
| 5    | <code>-rank(abs(zscore(turnover_rate)))</code>        | liquidity and crowdedness                                          | 5.668     | 9.484      |
| 6    | <code>-rank(abs(zscore(turnover_rate)-zscor...</code> | liquidity and crowdedness                                          | 5.547     | 5.024      |
| 7    | <code>-rank(abs(zscore(turnover_rate)-zscor...</code> | liquidity and crowdedness; risk-regime conditioning                | 5.410     | 0.085      |
| 8    | <code>-rank(abs(zscore(turnover_5d)-zscore(...</code> | liquidity and crowdedness; risk-regime conditioning                | 5.403     | 8.120      |
| 9    | <code>-rank(abs(ret_20d-abnormal_vol_rev))</code>     | medium-horizon continuation or reversal; risk-regime conditioning  | 5.103     | -1.073     |
| 10   | <code>-rank(abs(zscore(turnover_5d)-zscore(...</code> | liquidity and crowdedness; short-horizon overreaction correction   | 4.852     | 5.900      |

### 7.1 Child-factor economic dictionary

The following dictionary explains the detected leaf proxies used by the strict candidates and frozen greedy seed basket. Operators such as ranking, z-scoring, absolute deviation, and signed aggregation change scaling or interaction shape; they do not create a new economic primitive.

| Child proxy                       | Economic interpretation                                                      |
|-----------------------------------|------------------------------------------------------------------------------|
| <code>abnormal_vol_rev</code>     | abnormal-volume reversal; a short-lived price-pressure proxy                 |
| <code>amihud_20d</code>           | Amihud illiquidity; price impact per traded value                            |
| <code>log_dollar_vol</code>       | log trading-value scale; a liquidity, capacity, and crowding proxy           |
| <code>ret_120d_skip5</code>       | longer-horizon return excluding the most recent week; persistent trend proxy |
| <code>ret_20d</code>              | medium-horizon return; continuation versus reversal state                    |
| <code>turnover_5d</code>          | short-horizon turnover intensity; a transient liquidity-demand proxy         |
| <code>turnover_rate</code>        | share turnover intensity; a trading-pressure and crowdedness proxy           |
| <code>volume_profile_ratio</code> | relative volume-shape abnormality; an intraday participation proxy           |

## 8 Exploratory hierarchical nesting

Matrix-level nesting caches each OOS combo matrix and reuses it as a higher-level input. New records preserve lineage through `superalpha_ref(history-id)` references while loading the corresponding cached matrix. This is an engineering contribution and an exploratory modeling result: selecting the best lower layer after viewing OOS performance is not confirmatory evidence.

Exploratory hierarchical nesting ablation (best OOS Sharpe per layer)

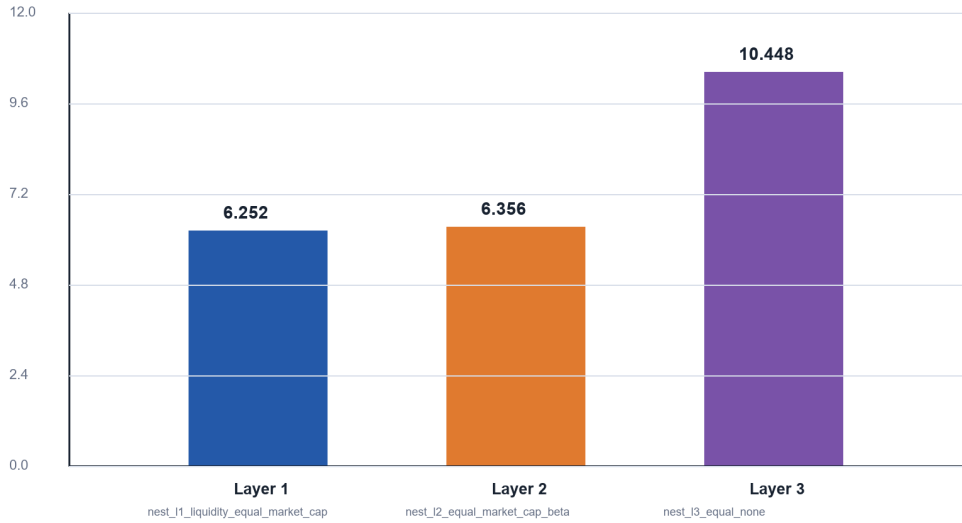


Figure 6: Exploratory L1–L3 nesting ablation.

## 9 Threats to validity

1. The OOS window contains one calendar year. A publishable asset-pricing claim requires multi-period walk-forward validation and regime splits.
2. Transaction costs, delay sensitivity, T+1 execution, price limits, and capacity remain to be integrated.
3. The factor search space is large. Multiple testing is controlled operationally by IS-only selection and frozen OOS reporting, but formal false-discovery analysis remains future work.
4. Historical cached combos created before lineage preservation remain valid matrices but cannot support leaf-level economic attribution. They are marked as legacy artifacts.
5. Industry neutralization is absent because reliable industry labels are unavailable in the current snapshot.

## 10 Reproducibility

The frozen evidence file is `experiment_results/final_audit_results.json`. The runner is `run_final_audit_experiments.py`. Every confirmatory row stores input IDs, weighting method, neutralization mode, requested and valid feature counts, skipped features, metrics, and PnL. The Flask service remains available for remote read-only review throughout the audit; external Alpha submission is outside the scope of this work.

## 11 Conclusion

The final audit replaces an earlier narrative report with a reproducible evidence chain. The main empirical lesson is not that arbitrary stacking guarantees high Sharpe. Rather, strict temporal separation, explicit neutralization, and true PnL-correlation constraints expose how quickly superficially strong variants collapse into the same economic bet. Matrix-level nesting remains useful, but only when its lineage and exploratory status are preserved.

## References

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